

GRANT COUNTY AP, WV, USA

WMO: 724165

Lat: 38.994N

Lon: 79.144W

Elev: 293

StdP: 97.86

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 03725

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.1	-9.1	-21.0	0.6	-7.7	-17.8	0.8	-4.7	11.7	4.1	10.7	3.2	1.5	280	0.416

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	33.8	22.4	32.5	22.2	31.3	21.8	25.3	30.4	24.4	29.4	23.5	28.3	2.8	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.5	27.3	22.9	18.2	26.4	22.3	17.6	25.8	79.2	30.0	75.2	29.9	71.7	28.9	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.0	6.7	-16.2	36.1	3.1	2.0	-18.5	37.5	-20.3	38.7	-22.0	39.8	-24.3	41.2		
(5)			-16.8	25.9	3.0	1.4	-18.9	26.9	-20.7	27.8	-22.3	28.6	-24.5	29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.9	1.6	2.7	7.0	12.8	17.6	21.9	23.8	23.1	19.5	13.5	7.5	3.4	(6)	
(7)		DBStd	9.13	6.05	5.59	5.53	4.86	4.37	3.02	2.74	2.68	3.66	4.63	4.59	4.84	(7)	
(8)		HDD10.0	971	268	213	129	26	2	0	0	0	0	18	101	213	(8)	
(9)		HDD18.3	2545	518	439	353	176	69	6	1	2	29	161	326	464	(9)	
(10)		CDD10.0	2032	8	8	35	110	237	356	427	406	286	127	26	7	(10)	
(11)		CDD18.3	564	0	0	1	9	46	112	169	149	65	12	0	0	(11)	
(12)		CDH23.3	5779	1	3	24	204	508	1123	1627	1399	705	173	12	2	(12)	
(13)		CDH26.7	2230	0	0	2	59	158	417	694	583	271	44	1	0	(13)	
(14)	Wind	WSAvg	1.6	2.2	2.3	2.3	2.1	1.3	1.2	1.0	0.8	0.9	1.4	1.5	1.8	(14)	
(15)	Precipitation	PrecAvg	1004	68	61	91	87	104	91	112	91	88	64	64	72	(15)	
(16)		PrecMax	1527	158	148	156	171	213	162	196	161	228	142	131	118	(16)	
(17)		PrecMin	825	26	18	22	40	42	26	37	47	3	9	17	36	(17)	
(18)		PrecStd	177	36	32	35	32	43	37	43	29	63	42	28	24	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.1	22.4	26.2	32.0	32.6	34.3	36.4	35.9	34.1	30.7	25.0	21.1	(19)	
(20)			MCWB	15.6	14.0	14.7	16.0	21.4	22.4	22.5	22.8	20.8	20.0	14.9	15.2	(20)	
(21)		2%	DB	17.7	18.1	22.8	28.0	30.6	32.6	33.7	33.2	32.1	27.6	22.2	17.4	(21)	
(22)			MCWB	12.4	11.3	13.5	16.4	20.9	22.6	22.7	23.1	20.3	18.1	14.2	11.9	(22)	
(23)		5%	DB	14.0	15.9	20.2	25.9	28.1	31.1	32.3	32.1	30.0	25.1	18.8	14.0	(23)	
(24)			MCWB	9.9	9.9	12.1	15.7	19.7	22.0	22.6	22.5	20.6	17.4	12.5	9.7	(24)	
(25)		10%	DB	11.1	12.4	17.3	22.7	26.4	29.0	30.9	30.1	27.6	22.3	16.4	11.3	(25)	
(26)			MCWB	7.5	7.3	11.1	14.4	19.0	21.3	22.4	21.8	19.5	16.3	10.7	7.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.2	15.5	17.1	20.7	24.4	26.4	26.7	26.4	24.4	22.4	17.4	15.8	(27)	
(28)			MCDWB	20.4	22.3	23.6	25.6	29.2	30.0	29.9	31.7	28.4	26.7	21.9	19.3	(28)	
(29)		2%	WB	13.5	12.7	15.1	18.2	22.7	24.6	25.3	25.1	23.0	20.5	15.3	12.8	(29)	
(30)			MCDWB	16.5	17.0	20.6	24.4	27.6	29.6	30.2	30.5	27.8	24.9	20.2	16.3	(30)	
(31)		5%	WB	10.2	10.2	13.5	16.9	21.4	23.4	24.4	24.0	22.1	18.9	13.5	10.7	(31)	
(32)			MCDWB	13.8	15.0	19.0	23.1	26.2	28.2	29.4	29.0	27.1	23.1	17.4	13.4	(32)	
(33)		10%	WB	7.5	7.6	11.4	15.4	20.1	22.4	23.5	23.0	20.9	17.1	11.7	8.0	(33)	
(34)			MCDWB	10.5	11.6	16.4	21.5	24.5	27.1	28.2	27.5	24.9	20.9	15.3	10.3	(34)	
(35)	Mean Daily Temperature Range			MDBR	10.6	11.4	12.4	14.3	13.4	13.1	13.0	13.1	13.9	13.6	13.1	10.6	(35)
(36)		5% DB	MCDBR	15.2	17.1	17.9	19.6	16.8	15.5	15.1	15.9	17.5	18.3	18.6	15.4	(36)	
(37)			MCWBR	10.3	11.0	10.0	9.5	8.2	7.0	5.9	6.5	7.7	9.7	11.4	10.6	(37)	
(38)		5% WB	MCDWB	14.1	15.7	15.6	16.0	14.1	13.4	12.8	13.0	13.7	14.7	15.0	14.0	(38)	
(39)			MCWBR	10.4	11.0	9.6	8.9	7.7	7.0	6.2	6.2	7.3	8.5	10.3	10.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.300	0.320	0.349	0.402	0.440	0.493	0.507	0.520	0.411	0.358	0.318	0.299	(40)	
(41)		taud		2.474	2.395	2.387	2.267	2.179	2.064	2.046	1.981	2.281	2.432	2.507	2.509	(41)	
(42)		Ebn at Noon		879	903	908	875	844	798	782	758	831	848	850	854	(42)	
(43)		Edh at Noon		79	98	110	132	147	165	166	173	121	94	76	72	(43)	
(44)	All-Sky Solar Radiation	RadAvg		1.89	2.71	3.69	4.76	5.26	5.83	5.65	5.13	4.26	3.16	2.25	1.66	(44)	
(45)		RadStd		0.18	0.37	0.24	0.40	0.47	0.41	0.34	0.27	0.52	0.37	0.22	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (43 neighbors)	+0.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+114	+61

Nomenclature: See separate page